Olga Kondratenko

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Professional Summary

Machine Learning Engineer with 4+ years of experience developing AI, scientific computing, and medical imaging software for healthcare applications. Experienced in Python-based software development, computer vision, deep learning, quantitative image analysis, and processing large-scale medical imaging datasets. Strong background in developing production-ready software, DICOM imaging workflows, healthcare data integration, and AI-assisted clinical solutions within regulated environments. M.S. in Computer Science with research experience in machine learning, data analysis, and scientific software development.

Technical Skills

Scientific Computing & Data Analysis: Python, C++, Java, JavaScript, SQL, NumPy, pandas, SciPy, Jupyter Notebook, Matplotlib, Seaborn, statistical analysis, exploratory data analysis (EDA), feature engineering, scientific visualization, data processing, quantitative analysis

Machine Learning & Artificial Intelligence: Deep Learning, Computer Vision, CNNs, RNNs, LSTMs, Transformers, PyTorch, TensorFlow, Keras, scikit-learn, Large Language Models (LLMs), Agentic AI, model training, validation, hyperparameter tuning, inference pipelines, AI workflow development

Medical Imaging: Medical image analysis, DICOM, Orthanc, OHIF Viewer, Sonador, 3D medical imaging, volumetric image processing, image segmentation, bone segmentation, medical imaging visualization, healthcare metadata, clinical imaging workflows

Software Engineering: REST APIs, Flask, Docker, Kubernetes, Git, GitHub, Postman, Apache Airflow, Kafka, unit testing, integration testing, automation testing, software architecture, production deployment

Operating Systems: Linux, Ubuntu, macOS, WSL2, Windows

Professional Experience

Software Engineer – Robotics R&D

Smith+Nephew | December 2022 – July 2026

- Developed production-grade AI and software solutions for medical imaging and healthcare applications using Python and JavaScript.
- Developed software for processing, visualization, and analysis of large-scale medical imaging and healthcare datasets.
- Developed deep learning and computer vision models for 3D medical image analysis, including bone segmentation, anatomical modeling, and image representation.
- Processed and curated multimodal healthcare datasets, including volumetric medical imaging, clinical metadata, and text data to support AI model development and validation.
- Designed AI-assisted clinical workflows integrating Large Language Models (LLMs) with healthcare metadata and imaging systems.
- Worked extensively with DICOM imaging data, Orthanc PACS, OHIF Viewer, and healthcare interoperability standards, including exposure to FHIR.

- Collaborated with multidisciplinary engineering and research teams to transition research prototypes into stable production systems.
- Developed automated testing frameworks, including unit, functional, and desktop automation tests, ensuring software reliability and reproducibility.
- Produced technical documentation supporting medical software development, validation, and regulatory compliance activities.

Machine Learning Engineer (Volunteer)

MLDL Solutions | May 2022 – December 2022

- Performed exploratory data analysis and feature engineering on large environmental and geospatial datasets using Python, pandas, and NumPy.
- Developed and evaluated machine learning and deep learning models using TensorFlow and PyTorch.
- Built data visualization dashboards using D3.js to communicate analytical results.
- Worked with Google Earth Engine and Azure cloud services to process large-scale datasets.

AI Researcher (Volunteer)

University of San Francisco – MAGICS Lab | September 2021 – December 2023

- Conducted research involving data preprocessing, feature engineering, machine learning model development, validation, and performance evaluation.
- Processed and analyzed large datasets using Python scientific computing libraries.
- Developed data visualization tools and analytical workflows supporting research activities.
- Trained, tested, and validated deep learning models for predictive analysis.

Teaching Assistant

University of San Francisco | January 2022 – May 2022

- Assisted students in software engineering and programming courses through office hours, technical mentoring, and assignment development.
- Developed automated unit tests to improve student feedback and code quality.

Systems Administration Assistant

University of San Francisco | January 2022 – May 2022

- Installed, configured, and maintained Linux, Windows, and macOS systems supporting university teaching and research laboratories.
- Diagnosed hardware and software issues, minimizing downtime for laboratory environments.

Education

University of San Francisco
Master of Science (M.S.), Computer Science
August 2020 – May 2022